



Office of the
Deputy Commissioner of
Maritime Affairs

THE REPUBLIC OF LIBERIA

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Marine Advisory 08/2026

Subject: Guidance for Safe Operation in Cold Weather and Low Temperatures

Reference:

- a) SOLAS Convention
- b) ISM Code
- c) [Marine Notice SAF-009](#)

Dear Shipowners/Operators/Recognized Organizations/Masters/Crew

Purpose

The purpose of this Marine Advisory is to provide guidance to shipowners, operators, recognized organizations, masters and crew for the safe operation of the ship, equipment, systems and machinery when operating/navigating in cold weather and low temperatures, following detention of a ship due to frozen pipes and damaged solenoid valve of the fixed local application fire-extinguishing system.

Background and Regulatory Framework

SOLAS Chapter I, Regulation 11(a) states that the condition of the ship and its equipment shall be maintained to conform with the provisions of the present regulations to ensure that the ship in all respects will remain fit to proceed to sea without danger to the ship or persons on board.

SOLAS Chapter II-2, regulation 14.2 requires at all times while a ship is in service, the fire protection systems and fire-fighting systems and appliances shall be maintained ready for use.

SOLAS Chapter III, Regulation 20.2 requires that all lifesaving appliances remain in working order and ready for immediate use at all times.

Operational readiness of the ship, its equipment, systems and machinery shall be maintained such that they are readily available in all expected weather conditions, including cold weather and low temperatures below the normal design temperature of the equipment, systems and machinery, and as a result of ice accretion from atmospheric icing or sea spray, freezing of liquid within a system or cold dense air affecting the running of machinery/draining of batteries.

SOLAS Chapter IX and the ISM Code:

- Section 1.2 requires the Company to provide for safe practices in ship operation and a safe working environment; assess all identified risks to its ships, personnel and the environment and

establish appropriate safeguards; and continuously improve safety management skills of personnel ashore and aboard ships.

- Section 7 requires the Company to establish procedures, plans and instructions, including checklists as appropriate, for key shipboard operations concerning the safety of the personnel, ship and protection of the environment. The various tasks should be defined and assigned to qualified personnel. This includes establishing procedures, plans and instructions, including checklists for operating in cold weather and low temperatures below normal design temperatures of equipment, systems and machinery.
- Section 10 requires the Company:
 - to establish procedures to ensure that the ship and its equipment are maintained in conformity with the provisions of the relevant rules and regulations and with any additional requirements which may be established by the Company;
 - to identify equipment and technical systems, the sudden operational failure of which may result in hazardous situations;
 - Safety Management System (SMS) to provide for specific measures aimed at promoting the reliability of such equipment or systems; and
 - To integrate the measures referred to above in the ship's operational maintenance routine.
- Section 6 requires the Company to ensure that all personnel involved in the Company's SMS have an adequate understanding of relevant rules, regulations, codes and guidelines; and establish and maintain procedures for identifying any training which may be required in support of the SMS and ensure that such training is provided for all personnel concerned.

Recommendation

Taking into consideration the above regulatory framework, shipowners/operators and Masters should implement the necessary procedures, checklists and training onboard prior to operating in cold weather and low temperatures to ensure the winterization and ready availability of ship's equipment, systems and machinery at all times; for which a non-exhaustive list of items to consider are:

1. Crew protection and personal protective equipment;
2. Freezing spray on exposed decks, hatch-covers, machinery and accommodation;
3. Movement on exposed decks, access arrangements to accommodation, survival craft, deck machinery, safety walkways;
4. Freezing ballast in exposed peak and top side tanks, ballast water treatment system including ballast pumps; stern tube cooling water tank;
5. Deck and hydraulic/electrical machinery, such as those for anchoring/mooring operations, cargo handling systems (cranes) and hatch covers: Availability, lubrication and pre-warming up;
6. Accommodation ladders, gangways, pilot ladders, mooring ropes;
7. Life-saving appliances: Survival craft, launching appliances, lifeboat equipment, lifeboat engine heating systems, fuel oil tanks, drinking water tanks, batteries, hydraulic motors/lines, lifebuoys;

8. Fire-fighting systems and appliances: Fire main, fixed local fire-extinguishing systems, fire-pumps (main and emergency), fixed deck foam systems, fixed gas fire-extinguishing systems, fire hoses, fire hydrants, fireman's outfits, fire-extinguishers;
9. Air horn, navigation lights, radar scanners, magnetic compass, gyro repeaters, VDR, navigation systems and equipment, emergency batteries, deck lighting, wheelhouse windows/heating/window-wash and wipers;
10. Airpipes, ventilators, dampers, P/V valves/breakers on tankers;
11. Main propulsion and auxiliary engines: Availability of cooling water from sea chests (refer to [MSC/Circ.504](#));
12. Steam to heating coils for various tanks, sea chests, other equipment (tracing steam);
13. Emergency generator and space heater;
14. Selection of lubricants, hydraulic oil, fuel oil, grease;
15. Propeller: Free of floating ice;
16. Accommodation heating and space heaters/insulation for critical equipment and appliances;
17. Availability of ice removal equipment/deicing salt, heating equipment and covers for deck machinery and equipment.

Manufacturer's equipment manuals should be consulted to determine the normal operational range of equipment, systems and machinery, and the scope and extent of winterization required.

Reference should be made to publications, such as '[Ice Navigation in Canadian Waters](#), **Ice Navigation Manual by BIMCO**, [ABS' Guide for Vessels Operating in Low Temperature Environments](#).

For more information, please contact Regulations and Standards Department at Regsandstandards@liscr.com or telephone +1 703 790 3434